

MATTHEW CASALI

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PROFESSIONAL SUMMARY

Innovative and creative data analyst with a strong background in sensing technology and deep learning model development. Experienced in applying advanced signal processing and machine learning classification techniques guided by integration of diverse data sources, high-fidelity acoustic modeling, and measurement for robust model training. Proven ability to deliver groundbreaking products, novel research, and impactful solutions for sponsors.

NOTABLE WORK

- Novel metamaterial-based beamforming acoustic transducer developed under funding from DARPA.
- Modeling and design of passive and active large aperture array beamforming for government sponsors.
- Patented ultrasonic method for non-destructive evaluation of composite materials for the U.S. Air Force.
- Focused ultrasound transmission techniques to reduce stress of nuclear waste canister welds for the DOE.
- Wireless acoustic sensor system for monitoring gas properties in DOE nuclear waste canisters.

PROFESSIONAL EXPERIENCE

Data Analyst and Acoustic Modeler

May 2024 - July 2025

Johns Hopkins University Applied Physics Laboratory

- Increased accuracy and reliability in performance modeling of cutting-edge beamforming sensor systems through the design of high-precision analytical and finite element methods.
- Innovated signal processing algorithms by leveraging machine learning and image processing techniques, enabling highly accurate signal detection and noise characterization.
- Streamlined modeling workflows and expanded the applicability of acoustic analysis methods across varied use cases by developing and maintaining object-oriented MATLAB toolboxes.
- Enhanced underwater acoustic simulations by integrating diverse data sources, improving predictive accuracy for real-world applications.

Research and Development Engineer

Sept. 2023 - Jan. 2024

X-Wave Innovations, Inc.

- Regularly presented innovative technical approaches and findings to multidisciplinary teams of engineers and scientists by translating complex acoustic data into actionable insights that guided decision-making.
- Authored technical reports and proposals for government sponsors, contributing to solutions across various acoustic modeling and data analysis domains.
- Designed and evaluated multiple test systems from scratch, including a force measurement apparatus and a vibrational analysis system for high power ultrasound.
- Improved the efficiency and accuracy of ultrasonic technology by enhancing power usage and focusing capabilities through the application of advanced optimization algorithms.

Data Consultant

Oct. 2020 - Dec. 2020

Conservation Metrics, Inc.

- Oversaw the organization and analysis of hundreds of hours of ultrasonic recordings of bat calls from over 20 National Forests in collaboration with the North American Bat Monitoring Program, providing valuable insights into bat population dynamics and conservation efforts.
- Informed precise action on the conservation of bat species through the synthesis of diverse data modalities into quantifiable and accessible reports on bat population statistics.

RESEARCH AND INTERNSHIPS

Graduate Research Assistant

Sept. 2021 - Aug. 2023

UT Austin, Applied Research Laboratories

- Pioneered new methods for acoustic sensing by measuring material interactions with sound in water, advancing the design of next-generation sensing systems.
- Developed composite materials that allow precise measurement of sound pressure and direction at a single point lowering the cost and increasing the accuracy of acoustic sensing systems.
- Enhanced the sensing capabilities of materials by applying advanced optimization techniques to the modeling of custom sensor components.

Research Intern

Jan. 2020 - Apr. 2020

UC Santa Cruz Fort Ord Natural Reserve

- Proposal and execution of a unique and novel case of environmental monitoring, resulting in a project that is currently used for educational outreach by the natural reserve.
- Developed a Python program to identify hummingbird calls and map their flight paths using data from a custom-built 3D microphone system.

EDUCATION

The University of Texas at Austin

2023

Master of Science, Mechanical Engineering (Acoustics)

University of California - Santa Cruz

2020

Bachelor of Science, Applied Physics

AWARDS AND APPOINTMENTS

- **Fellow, Cockrell School of Engineering.** Awarded in recognition of exemplary academic performance in pursuit of a Bachelor's degree and strength of application materials.
- **Secretary, Acoustical Society of America, Austin Student Chapter.** Tasked with social events, outreach, and community engagement.
- **Member, College Scholars Program – UC Santa Cruz.** Awarded in recognition of exemplary academic performance in high school and first-quarter courses.

SELECTED PUBLICATIONS AND PRESENTATIONS

- **Matthew A. Casali**, "Material property extraction procedure for electromomentum coupled meta-materials." Master's Thesis, 2023.
- Samuel P. Wallen, **Matthew A. Casali**, Benjamin M. Goldsberry, and Michael R. Haberman, "Polarizability of electromomentum coupled scatterers", Proc. Mtgs. Acoust. 46, 065002 (2022) <https://doi.org/10.1121/2.0001597>.
- **Matthew A. Casali**, Andrew R. McNeese, Samuel P. Wallen, Michael R. Haberman (2022) "Impedance tube method for characterization of one-dimensional electro-momentum coupled materials" Presented at the 182nd Meeting of the Acoustical Society of America, Nashville, TN.